

Islamic University of Madinah
Faculty of Science
Mathematics Department



Calculus II (MATH 3112)

Course References:

- ◇ J. Stewart, Calculus: Early Transcendentals; Publisher, Cengage Learning; 8th edition (February 4, 2015); Language, English; Hardcover, 1368 pages; ISBN-10, 1285741552.
- ◇ Saudi Digital Library: <https://sdl.edu.sa/SDLPortal/ar/Publishers.aspx>

CHAPTER 5: INTEGRALS

Section 5.1: Area Problem

Section 5.2: The Definite Integral (Definition)

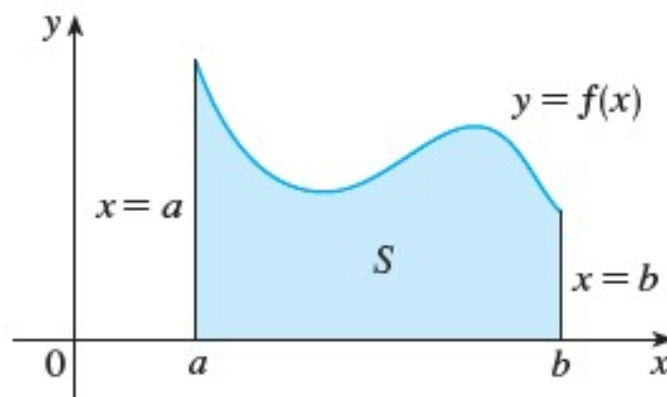
●● **Course Materials will be covered from:** Course Text Book and These Lecture Notes

Section 5.1 (Area Problem)

OBJECTIVES:

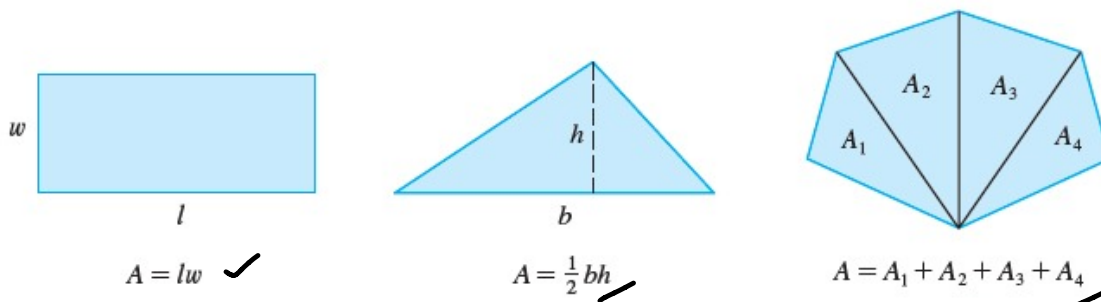
- Find (upper and lower) estimates of areas under curves using (left and right) endpoints or any sample points.
- Define the area under the graph of a continuous function as the limiting value of the sum of the areas of the approximating rectangles.
- Be able to use the Sigma notation to write the sum of terms

We begin by attempting to solve the area problem: Find the area of the region S that lies under the curve $y = f(x)$ from a to b . This means that S , illustrated in Figure 1, is bounded by the graph of a continuous function f [where $f(x) > 0$], the vertical lines $x = a$ and $x = b$, and the x -axis.

**FIGURE 1**

$$S = \{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x)\}$$

In trying to solve the area problem we have to ask ourselves: What is the meaning of the word area? This question is easy to answer for regions with straight sides. For a rectangle, the area is defined as the product of the length and the width. The area of a triangle is half the base times the height. The area of a polygon is found by dividing it into triangles (as in Figure 2) and adding the areas of the triangles.

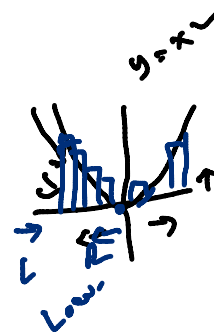


However, it is not so easy to find the area of a region with curved sides. We all have an intuitive idea of what the area of a region is. But part of the area problem is to make this intuitive idea precise by giving an exact definition of area. Recall that in defining a tangent we first approximated the

Remarks:

To divide the interval $[a, b]$ to n -equal subintervals, then $\Delta x = \frac{b-a}{n}$.

- If f is an increasing function, then R.E.P gives Over estimation.
- If f is an increasing function, then L.E.P gives Lower estimation.
- If f is an decreasing function, then L.E.P gives Over estimation.
- If f is an decreasing function, then R.E.P gives Lower estimation.



Definition 1. The area A of the region S that lies under the graph of the continuous function $f(x)$ is the limit of the sum of the areas of approximating rectangles:

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta x f(x_i) = \lim_{n \rightarrow \infty} [\Delta x f(x_1) + \Delta x f(x_2) + \Delta x f(x_3) + \dots + \Delta x f(x_n)].$$

//

Ex. 1, P. 366: Let R be the region under the graph of $f(x) = x^2$, $x \in [1, 2]$. Estimate the area using 4-equal subintervals and right endpoints. Sketch the graph and the rectangles. Is your estimate an underestimate or an overestimate?

$$\begin{aligned}
 A &= \sum_{i=1}^n f(x_i) \Delta x \\
 &= \frac{1}{4} [f(1.25) + f(1.5) + f(1.75) + f(2)] \\
 &= \frac{1}{4} (10.875) \\
 &= \frac{87}{32} = 2.718
 \end{aligned}$$

and we are considering $\downarrow$ so it's

f is increasing on $[1, 2]$, right end points over estimate

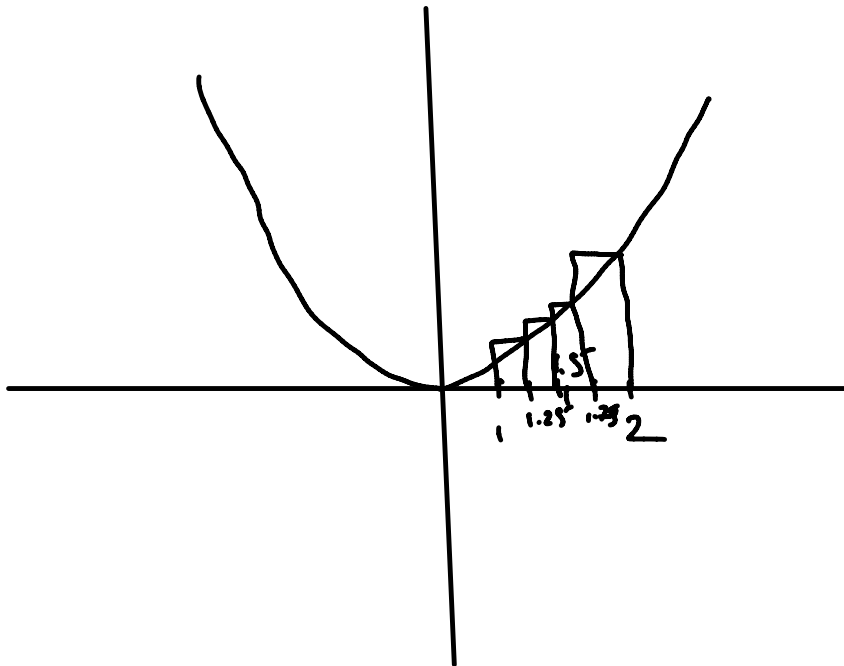
$$\begin{aligned}
 \Delta x &= \frac{b-a}{n} = \frac{2-1}{4} \\
 &= \frac{1}{4} = 0.25
 \end{aligned}$$

$$x_1 = 1 + 0.25 = 1.25$$

$$x_2 = 1.5$$

$$x_3 = 1.75$$

$$x_4 = 2$$



Ex.2: Let R be the region under the graph of $f(x) = x^2$, $x \in [-2, -1]$. Estimate the area using 4-equal subintervals and left endpoints. Sketch the graph and the rectangles. Is your estimate an underestimate or an overestimate?

$$A = \sum_{i=1}^4 \Delta x f(x_i)$$

$$= \frac{1}{4} [f(-2) + f(-1.75) + f(-1.5) + f(-1.25)]$$

$$= \frac{1}{4} (10.875)$$

$$= \frac{87}{32}$$

$$\Delta x = \frac{(-1) - (-2)}{4}$$

$$= \frac{1}{4}$$

$$x_1 = -2$$

$$x_2 = -1.75$$

$$x_3 = -1.75 + 0.25$$

$$= -1.5$$

$$x_4 = -1.25$$



(overestimate)

Ex.3 (Exercise 4, P. 375): Estimate the area of the region under the curve $y = \sin x$, $x \in [0, \frac{\pi}{2}]$ using 4-equal sub intervals and:

i) Right end points.

ii) Left end points.

$$(i) A_R = \sum_{i=1}^4 \Delta x f(x_i)$$

$$= \frac{\pi}{8} [f(\frac{\pi}{2}) + f(\frac{3\pi}{8}) + f(\frac{\pi}{4}) + f(\frac{\pi}{8})]$$

$$= 1.1834$$

$$(ii) A_L = \sum_{i=1}^4 \Delta x f(x_i)$$

$$= \frac{\pi}{8} [f(0) + f(\frac{3\pi}{8}) + f(\frac{\pi}{4}) + f(\frac{\pi}{8})]$$

$$= 0.790$$

Ask your lecturer whether you can choose

$$\Delta x = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8}$$

left:

$$x_1 = 0$$

$$x_2 = \frac{\pi}{8}$$

$$x_3 = \frac{\pi}{4}$$

$$x_4 = \frac{3\pi}{8}$$

Right:

$$x_1 = \frac{\pi}{8}$$

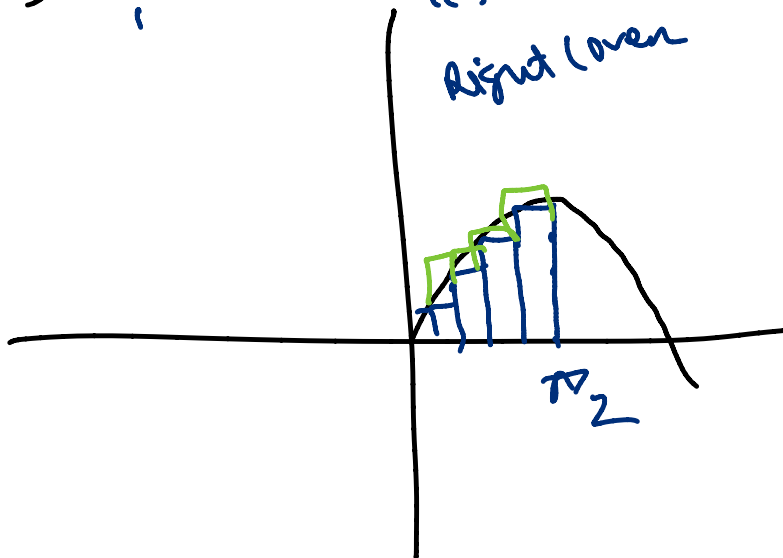
$$x_2 = \frac{3\pi}{8}$$

$$x_3 = \frac{\pi}{4}$$

$$x_4 = \frac{\pi}{8}$$

left (under estimate)

Right (over)



Ex.4: Find an upper estimate to the area of the region under the curve $y = \cos x$, $x \in [0, \frac{\pi}{2}]$ using 3-equal sub intervals.

$$i) A_L = \sum_{i=1}^n \Delta x f(x_i)$$

$$= \pi/6 [f(0) + f(\pi/6) + f(\pi/3)]$$

$$= 1.2389$$

$$\Delta x = \frac{\pi/2 - 0}{3}$$

$$= \pi/6$$

$$x_1 = 0$$

$$x_2 = \pi/6$$

$$x_3 = \pi/3$$

Ex.5: Find a lower estimate to the area of the region under the curve $y = e^x$, $x \in [0, 3]$ using 6-equal sub intervals.

$$A_L = \sum_{i=1}^6 \Delta x e^{x_{i-1}}$$

$$= \frac{1}{2} [e^0 + e^{0.5} + e^1 + e^{1.5} + e^2 + e^{2.5}]$$

$$= 14.7101$$

(Ans)

✗

$$\Delta x = \frac{3-0}{6}$$

$$= 0.5$$

$$x_1 = 0$$

$$x_2 = 0.5$$

$$x_3 = 1$$

$$x_4 = 1.5$$

$$x_5 = 2$$

$$x_6 = 2.5$$

Ex.6 (Exercise 3, P. 375): Estimate the area of the region under the curve $f(x) = \frac{1}{x}$, $x \in [1, 2]$ using 4-equal sub intervals and:

i) Right end points.

ii) Left end points.

Left

$$A_L = \frac{1}{4} [f(1) + f(1.25) + f(1.5) + f(1.75)]$$

$$= \frac{1}{4} \left(\frac{4}{9} + \frac{2}{3} + \frac{4}{5} + \frac{1}{2} \right)$$

$$= 0.6345$$

(underestimate)

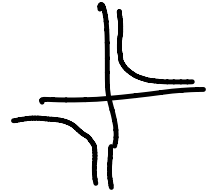
Right

$$A_R = \frac{1}{4} [f(2) + f(1.75) + f(1.5) + f(1.25)]$$

$$= \frac{1}{4} \left(\frac{4}{5} + \frac{4}{9} + \frac{2}{3} + 1 \right)$$

$$= 0.750$$

→ overestimated



$$\Delta x = \frac{2-1}{4}$$

$$\Delta x = 0.25$$

$$x_1 = 1$$

$$x_2 = 1.25$$

$$x_3 = 1.5$$

$$x_4 = 1.75$$

Right

$$x_1 = 1.25$$

$$x_2 = 1.5$$

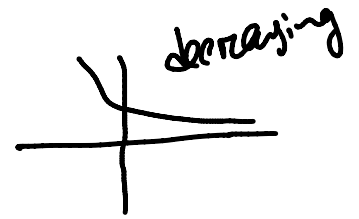
$$x_3 = 1.75$$

$$x_4 = 2$$

Ex.7: Find a lower estimate to the area of the region under the curve $y = e^{-x}$, $x \in [0, 3]$ using 6-equal sub intervals.

$$\Delta x = \frac{3-0}{6} = \frac{1}{2}$$

lower estimate decreasing RHP



$$A_R = \sum_{i=1}^6 \Delta x f(x_i)$$

$$= \Delta x (f(\frac{1}{2}) + f(1) + f(1.5) + f(2) + f(2.5) + f(3))$$

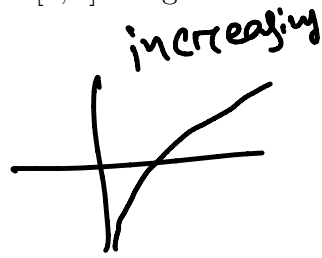
$$= \frac{1}{2} (e^{-0.5} + e^{-1} + e^{-1.5} + e^{-2} + e^{-2.5} + e^{-3})$$

$$= 0.7323 \text{ (Ans)}$$

Ex.8: Find a lower estimate to the area of the region under the curve $y = \ln x$, $x \in [2, 4]$ using 4-equal sub intervals.

$$\Delta x = \frac{4-2}{4} = \frac{1}{2}$$

increasing + lower estimate LEP



$$A_L = \sum_{i=1}^4 \Delta x f(x_i)$$

$$\therefore \frac{1}{2} [f(2) + f(2.5) + f(3) + f(3.5)]$$

$$= 1.98 \text{ (underestimate)}$$

Section 5.2: The Definite Integral (Definition)

OBJECTIVES:

- Define the definite integral of a continuous function and interpret it in terms of areas above and below the x-axis.
- Evaluate the definite integral of simple functions using the definition and the sum rules.

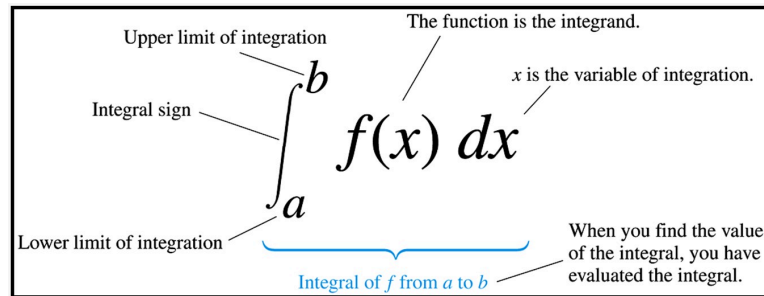
Theorem 1 If f is continuous on $[a, b]$, then f is integrable and

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{i=1}^n f\left(a + \left(\frac{b-a}{n}\right)i\right).$$

Summation Rules: Let n, m be natural numbers and let c be a constant. Then

$$\sum_{i=m}^n (x_i \pm y_i) = \sum_{i=m}^n x_i \pm \sum_{i=m}^n y_i, \quad \sum_{i=m}^n cx_i = c \sum_{i=m}^n x_i, \quad \sum_{i=1}^n C = nC$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^{\infty} i^3 = \left[\frac{n(n+1)}{2} \right]^2$$



Properties of the Integral:

- $\int_a^a f(x) dx = 0.$
- $\int_a^b f(x) dx = - \int_b^a f(x) dx.$
- $\int_a^b c dx = c(b-a),$ where c is any constant
- $\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$
- $\int_a^b cf(x) dx = c \int_a^b f(x) dx,$ where c is any constant
- $\int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$

Ex.1 (Example 2, P. 381): Evaluate the following integral by interpreting the integral as a limit of a Riemann sum:

$$\int_0^3 (x^3 - 6x) dx.$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \Delta x \sum_{i=1}^n f(x_i)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(\frac{3}{n} i \right)^3 - 6 \left(\frac{3}{n} i \right) \right] \left(\frac{3}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(\frac{27i^3}{n^3} \right) - \frac{18i}{n} \right] \left(\frac{3}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{81i^3}{n^4} - \frac{54i}{n^2} \right]$$

$$= \lim_{n \rightarrow \infty} \left(\frac{81}{n^4} \sum_{i=1}^n i^3 - \frac{54}{n^2} \sum_{i=1}^n i \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{81}{n^4} \left(\frac{n(n+1)}{2} \right)^2 - \frac{54}{n^2} \cdot \frac{n(n+1)}{2} \right)$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{n^4} \cdot \frac{n^2(n+1)^2}{4} - \frac{54}{n^2} \cdot \frac{n^2+n}{2} \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{n^4} \cdot \frac{n^2(n^2+2n+1)}{4} - \frac{54}{2} \cdot \frac{n^2+n}{n^2} \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{4} \cdot \frac{n^4+2n^3+1}{n^4} - \frac{54}{2} \left(1 + \frac{1}{n} \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{4} \cdot \left(1 + \frac{2}{n} + \frac{1}{n^4} \right) - \frac{54}{2} \left(1 + \frac{1}{n} \right) \right\}$$

$$= \frac{81}{4} - 27 = -\frac{27}{4} \text{ (Ans)}$$

$$\Delta x = \frac{b-a}{n}$$

$$= \frac{3-0}{n} = \frac{3}{n}$$

$$x_i = a + \left(\frac{b-a}{n} \right) i$$

$$= 0 + \left(\frac{3}{n} \right) i$$

$$= \frac{3}{n} i$$

Ex.2: Evaluate the following integral by interpreting the integral as a **limit** of a Riemann sum:

$$\Delta x = \frac{5-1}{n} \quad \text{at } \frac{5-1}{n} \cdot i \quad \int_1^5 (2x+3) dx. \quad = 1 + \frac{5-1}{n} \cdot i$$

21.

$$x_i = 1 + \frac{4}{n}i$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left\{ \left(2 \left(1 + \frac{4}{n}i \right) + 3 \right) \left(\frac{4}{n} \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left\{ \left(2 + \frac{8}{n}i + 3 \right) \left(\frac{4}{n} \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left\{ \left(5 + \frac{8}{n}i \right) \left(\frac{4}{n} \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{20}{n} + \frac{32}{n^2}i \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{20}{n} \sum_{i=1}^n 1 + \frac{32}{n^2} \cdot \sum_{i=1}^n i \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{20}{n} \cdot n + \frac{32}{n^2} \cdot \frac{n(n+1)}{2} \right)$$

$$= \lim_{n \rightarrow \infty} \left(20 + \frac{32}{2} \cdot \frac{n^2+n}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \left(20 + 16 \cdot \left(1 + \frac{1}{n} \right) \right) = 20 + 16 = 36$$

Ex.3 (Exercise 23, P. 389): Evaluate the following integral by interpreting the integral as a limit of a Riemann sum:

$$\int_{-2}^0 (x^2 + x) dx.$$

$$\begin{aligned} b-a &= 0 - (-2) \\ \Delta x &= \frac{0 - (-2)}{n} \\ &= \frac{2}{n} \\ x_i &= -2 + \frac{2}{n} \cdot i \end{aligned}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(-2 + \frac{2}{n} \cdot i \right)^2 + \left(-2 + \frac{2}{n} \cdot i \right) \right] \frac{2}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(4 - \frac{8i}{n} + \frac{4}{n^2} i^2 \right) + \left(-2 + \frac{2i}{n} \right) \right] \left(\frac{2}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left\{ \left(\frac{8}{n} - \frac{16i}{n^2} + \frac{8i^2}{n^3} \right) + \left(-\frac{4}{n} + \frac{4i}{n^2} \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{4}{n} - \frac{12i}{n^2} + \frac{8i^2}{n^3} \right)$$

$$= \lim_{n \rightarrow \infty} \left[\frac{4}{n} \sum_{i=1}^n 1 - \frac{12}{n^2} \sum_{i=1}^n i + \frac{8}{n^3} \sum_{i=1}^n i^2 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{4}{n} \cdot n - \frac{12}{n^2} \cdot \left(\frac{n(n+1)}{2} \right) + \frac{8}{n^3} \cdot \left(\frac{n(n+1)(2n+1)}{6} \right) \right]$$

$$= 4 - 6 + \frac{8}{3} = \boxed{\frac{2}{3}}$$

Ex.4: Evaluate the following integral by interpreting the integral as a **limit** of a Riemann sum:

$$\int_0^2 (x^3 - 1) dx.$$

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$x_i = a + \Delta x i$$

$$= 0 + \left(\frac{2}{n}\right)i$$

$$= \left(\frac{2}{n}\right)i$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(\left(\frac{2}{n}\right)i \right)^3 - 1 \right] \frac{2}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{8}{n^3} i^3 - 1 \right] \frac{2}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{16}{n^4} i^3 - \frac{2}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{16}{n^4} \sum_{i=1}^n i^3 - \frac{2}{n} \sum_{i=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{16}{n^4} \cdot \frac{n^2(n+1)^2}{4} - \frac{2}{n} \cdot n \right]$$

$$= \lim_{n \rightarrow \infty} \left[4 \cdot \frac{(n^2+n)(2n+1)}{n^4} - 2 \right]$$

$$= \lim_{n \rightarrow \infty} [4 - 2]$$

$$= 2$$

Ex.3 (Exercise 27, P. 389): Evaluate the following integral by interpreting the integral as a limit of a Riemann sum:

$$\int_a^b x \, dx.$$

$$\begin{aligned}\Delta x &= \frac{b-a}{n} \\ x_i &= a + \Delta x \cdot i \\ &= a + \left(\frac{b-a}{n}\right) \cdot i\end{aligned}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \cdot \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left\{ a + \left(\frac{b-a}{n}\right) i \right\} \cdot \frac{b-a}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{a(b-a)}{n} + \left(\frac{b-a}{n}\right)^2 i$$

$$= \lim_{n \rightarrow \infty} \frac{a(b-a)}{n} \cdot \sum_{i=1}^n 1 + \frac{(b-a)^2}{n^2} \cdot \sum_{i=1}^n i$$

$$= \lim_{n \rightarrow \infty} \frac{a(b-a)}{n} \cdot n + \frac{(b-a)^2}{n^2} \cdot \frac{n \cdot (n+1)}{2}$$

$$= \lim_{n \rightarrow \infty} a(b-a) + \frac{(b-a)^2}{2} \cdot \frac{n^2 + n}{n^2}$$

$$= \lim_{n \rightarrow \infty} a(b-a) + \frac{(b-a)^2}{2} \cdot \left(1 + \frac{1}{n}\right)$$

$$= a(b-a) + \frac{(b-a)^2}{2}$$